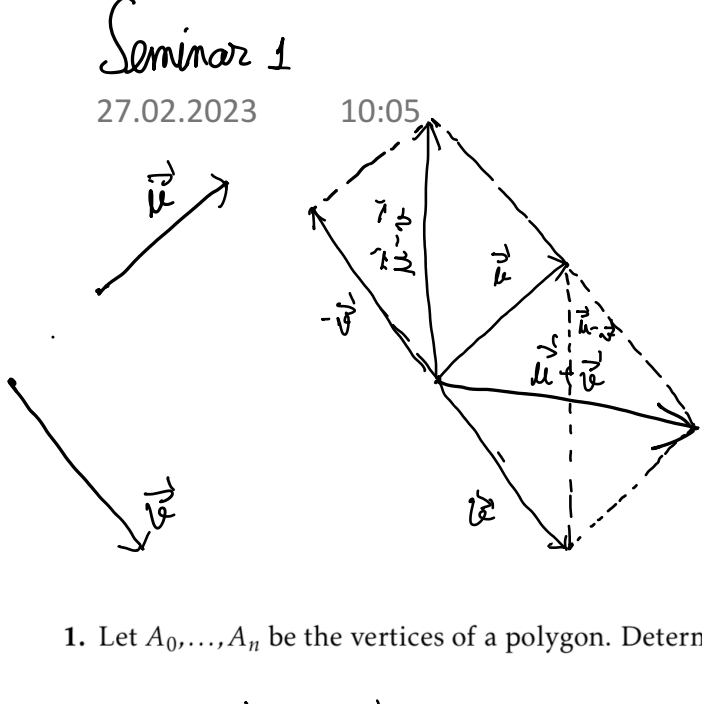
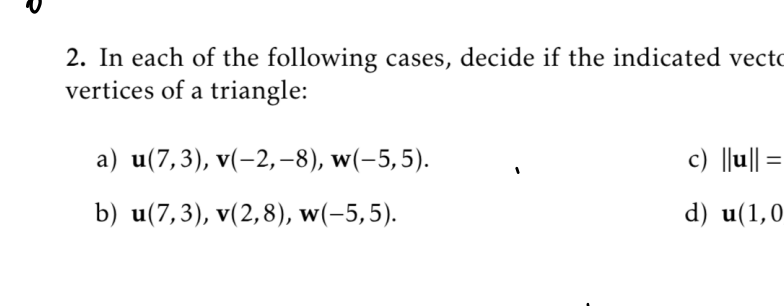




1. Let $A_0, \dots, A_n$ be the vertices of a polygon. Determine $\overrightarrow{A_0A_1} + \overrightarrow{A_1A_2} + \dots + \overrightarrow{A_{n-1}A_n} + \overrightarrow{A_nA_0}$.

$$\begin{aligned}\overrightarrow{A_0A_1} + \overrightarrow{A_1A_2} &= \overrightarrow{A_0A_2} \\ \overrightarrow{A_0A_2} + \overrightarrow{A_2A_3} &= \overrightarrow{A_0A_3} \\ &\vdots \\ \overrightarrow{A_0A_n} + \overrightarrow{A_nA_0} &= \vec{0}\end{aligned}$$



$$\vec{OH} = \frac{\vec{OA} + \vec{OB}}{2}$$

2. In each of the following cases, decide if the indicated vectors u, v, w can be represented with the vertices of a triangle:

- a) $u(7, 3), v(-2, -8), w(-5, 5)$. c) $\|u\| = 7, \|v\| = 3, \|w\| = 11$.
b) $u(7, 3), v(2, 8), w(-5, 5)$. d) $u(1, 0, 1), v(0, 1, 0), w(2, 2, 2)$.

a) $\vec{u}(7, 3), \vec{v}(-2, -8), \vec{w}(-5, 5)$
 $\vec{u} + \vec{v} + \vec{w} = (7, 3) + (-2, -8) + (-5, 5) = (0, 0)$ $\Rightarrow$ The triangle exists

b) $\vec{u}(7, 3), \vec{v}(2, 8), \vec{w}(-5, 5)$
 $\vec{u} - \vec{v} + \vec{w} = (7, 3) - (2, 8) + (-5, 5) = (0, 0)$ $\Rightarrow$ The triangle exists

c) $\|u\| = 7, \|v\| = 3, \|w\| = 11$
 $\|u\| + \|v\| = 10$
 $\|w\| = 11$ $\Rightarrow \|u\| + \|v\| < \|w\| \Rightarrow$ The triangle does not exist

d) $\vec{u}(1, 0, 1), \vec{v}(0, 1, 0), \vec{w}(2, 2, 2)$
It can not be represented
No matter the sign of $\vec{u}, \vec{v}$ and $\vec{w}$

$$\begin{cases} 0 + 1 \neq 2 \\ 0 - 1 \neq 2 \\ 0 + 1 \neq 2 \\ 0 - 1 \neq 2 \end{cases}$$

The triangle does not exist

4. Let $ABCD$ be a quadrilateral. Let M, N, P, Q be the midpoints of $[AB], [BC], [CD]$ and $[DA]$ respectively. Show that

$$\overrightarrow{MN} + \overrightarrow{PQ} = \vec{0}.$$

Deduce that the midpoints of the sides of an arbitrary quadrilateral form a parallelogram.

$$\begin{aligned}\vec{MN} &= \vec{MB} + \vec{BN} = \frac{1}{2}(\vec{AB} + \vec{BC}) = \frac{1}{2}\vec{AC} \\ \vec{PQ} &= \vec{PD} + \vec{DQ} = \frac{1}{2}(\vec{CD} + \vec{DA}) = \frac{1}{2}\vec{CA} \\ \vec{MN} + \vec{PQ} &= \frac{1}{2}(\vec{AC} + \vec{CA}) = \vec{0}\end{aligned}$$

$$\begin{aligned}\vec{MN} + \vec{PQ} &= \vec{0} \\ \vec{MN} &= -\vec{PQ} \\ M \neq Q \\ N \neq P\end{aligned} \Rightarrow MNPQ \text{ parallelogram}$$

7. Let $k = \frac{|CA|}{|CB|}$ be the ratio in which the point $C \in [AB]$ divides the segment $[AB]$. Show that for any point O we have

$$\overrightarrow{OC} = \frac{1}{1+k}(\overrightarrow{OA} + k\overrightarrow{OB}).$$

Deduce that C has coordinates

$$\left(\frac{a_1 + kb_1}{1+k}, \frac{a_2 + kb_2}{1+k}, \dots, \frac{a_n + kb_n}{1+k} \right)$$

where $A = A(a_1, \dots, a_n)$ and $B = B(b_1, \dots, b_n)$.

$$\begin{aligned}h &= \frac{CA}{CB} \Rightarrow \frac{|CB|}{|CA|} = \frac{1}{h} \\ \frac{|CB|}{|CA|} &= \frac{1}{h+1} \\ \frac{|CB|}{|AB|} &= \frac{1}{h+1} \\ \vec{CB} &= \frac{1}{h+1} \vec{AB} \\ \vec{BC} &= \frac{1}{h+1} \vec{BA}\end{aligned}$$

$$\vec{OC} = \vec{OB} + \vec{BC}$$

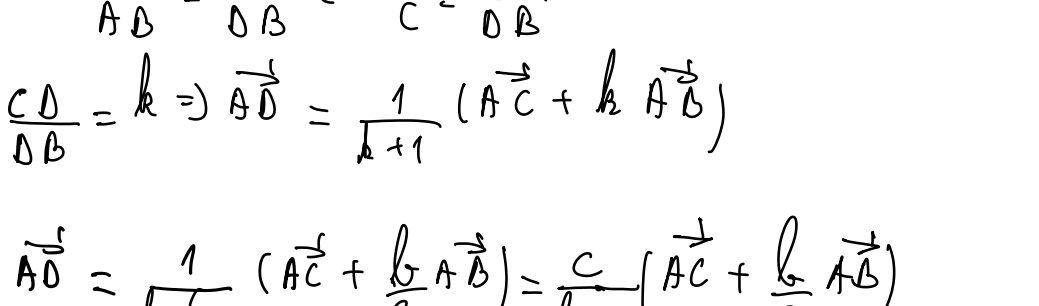
$$\vec{OC} = \vec{OB} + \frac{1}{h+1}(\vec{BO} + \vec{OA})$$

$$\vec{OC} = \frac{1}{h+1}(\vec{OA} + h\vec{OB})$$

$$C = \frac{1}{h+1}(A + hB)$$

$$C = \frac{A + hB}{h+1} \Rightarrow \begin{cases} C_1 = \frac{a_1 + hb_1}{h+1} \\ C_2 = \frac{a_2 + hb_2}{h+1} \\ \vdots \\ C_n = \frac{a_n + hb_n}{h+1} \end{cases}$$

8. Let ABC be a triangle and let $D \in [BC]$ be such that AD is an angle bisector. Express $\overrightarrow{AD}$ in terms of $b = \overrightarrow{AB}$ and $c = \overrightarrow{AC}$.



$$\frac{AC}{AB} = \frac{CD}{DB} \Leftrightarrow \frac{b}{c} = \frac{CD}{DB}$$

$$\frac{CD}{DB} = k \Rightarrow \vec{AD} = \frac{1}{k+1}(h\vec{c} + k\vec{b})$$

$$\vec{AD} = \frac{1}{\frac{b}{c}+1}(h\vec{c} + \frac{b}{c}h\vec{b}) = \frac{c}{b+c}(\frac{1}{h}\vec{b} + \frac{b}{c}h\vec{a})$$

3. Let $ABCDEF$ be a regular hexagon centered at O .

a) Express the vectors $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}, \overrightarrow{OD}$ in terms of $\overrightarrow{OE}$ and $\overrightarrow{OF}$.

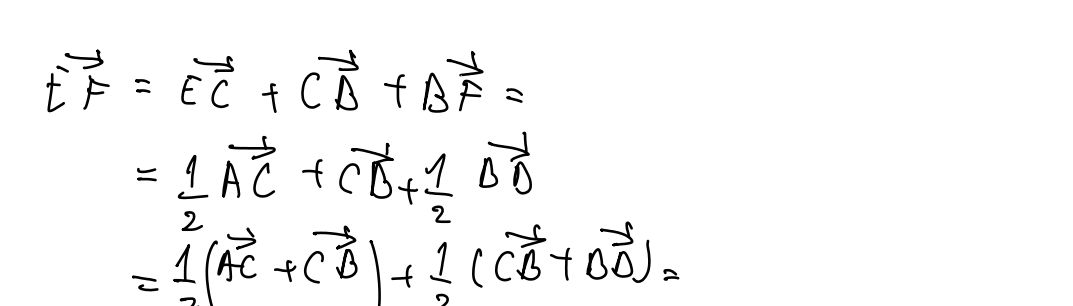
b) Show that $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = 3\overrightarrow{AD}$.

$$\begin{aligned}\vec{OA} &= \vec{OF} = -\vec{OE} + \vec{OF} \\ \vec{OB} &= \vec{EO} \\ \vec{OC} &= \vec{FO} \\ \vec{OD} &= \vec{OE} - \vec{OF}\end{aligned}$$

$$\begin{aligned}\vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} &= \vec{AO} \\ \vec{AC} + \vec{AF} &= \vec{AO}\end{aligned} \Rightarrow \vec{AB} + \vec{AC} + \vec{AD} + \vec{AE} + \vec{AF} = 3\vec{AO}$$

5. Let $ABCD$ be a quadrilateral. Let E be the midpoint of $[AC]$ and let F be the midpoint of $[BD]$. Show that

$$\overrightarrow{EF} = \frac{1}{2}(\overrightarrow{AB} + \overrightarrow{CD}) = \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{CB}).$$

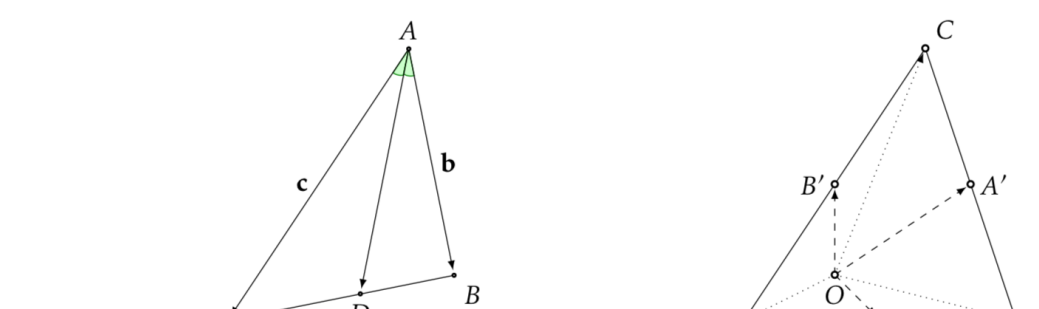


$$\begin{aligned}\vec{EF} &= \vec{EC} + \vec{CF} + \vec{FD} + \vec{DE} \\ &= \frac{1}{2}\vec{AC} + \frac{1}{2}\vec{BD} \\ &= \frac{1}{2}(\vec{AC} + \vec{BD}) + \frac{1}{2}(\vec{CD} + \vec{AB}) \\ &= \frac{1}{2}(\vec{AB} + \vec{CD}) \\ &= \frac{1}{2}(\vec{AD} + \vec{CB})\end{aligned}$$

6. Let $ABCD$ be a quadrilateral. Let E be the midpoint of $[AB]$ and let F be the midpoint of $[CD]$. Show that

$$\overrightarrow{EF} = \frac{1}{2}(\overrightarrow{AD} + \overrightarrow{BC}).$$

Deduce that the length of the midsegment in a trapezoid is the arithmetic mean of the lengths of the bases.

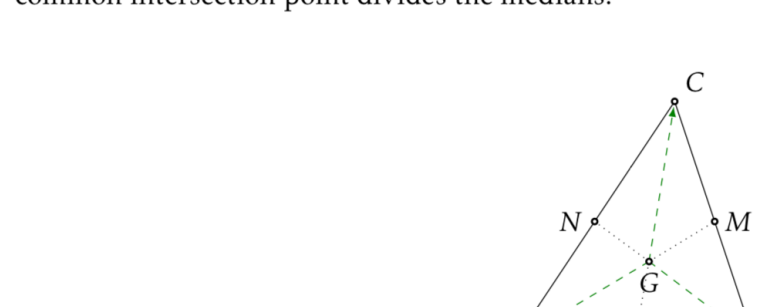


9. Let A', B' and C' be midpoints of the sides of a triangle ABC . Show that for any point O we have

$$\overrightarrow{OA'} + \overrightarrow{OB'} + \overrightarrow{OC'} = \overrightarrow{OA} + \overrightarrow{OB} + \overrightarrow{OC}.$$

$$\begin{aligned}\vec{OA'} &= \frac{1}{2}(\vec{OC} + \vec{OB}) \\ \vec{OB'} &= \frac{1}{2}(\vec{OA} + \vec{OC}) \\ \vec{OC'} &= \frac{1}{2}(\vec{OA} + \vec{OB})\end{aligned} \Rightarrow \vec{OA'} + \vec{OB'} + \vec{OC'} = \vec{OA} + \vec{OB} + \vec{OC}$$

10. Show that the medians in a triangle intersect in one point and deduce the ratio in which the common intersection point divides the medians.



$$\begin{aligned}\frac{CN}{NA} &= 1 \\ \frac{AP}{PB} &= 1\end{aligned} \Rightarrow \text{Ceva's converse} \Rightarrow CA \cap AB \cap BC = \{G\}$$

11. In each of the following cases, decide if the given points are collinear:

- a) $P(3, -5), Q(-1, 2), R(-5, 9)$. c) $P(1, 0, -1), Q(0, -1, 2), R(-1, -2, 5)$.
b) $A(1, 2), B(1, -3), C(3, 13)$. d) $A(-1, -1, -4), B(1, 1, 0), C(2, 2, 2)$.

a) $m_{PQ} = \frac{2+5}{-1-3} = -\frac{7}{4}$
 $m_{QR} = \frac{9-2}{-5-1} = -\frac{7}{4}$ $\Rightarrow P, Q, R$ collinear

b) $m_{AB} = \frac{-5}{-1-0} = \frac{1}{2}$
 $m_{BC} = \frac{1-6}{3-0} = \frac{5}{3}$ $\Rightarrow A, B, C$ not collinear

c) $d_{PQ} + d_{QR} = d_{PR}$

d) La fel

12. Let $ABCD$ be a tetrahedron. Determine the sums

- a) $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$, b) $\overrightarrow{AD} + \overrightarrow{BC} + \overrightarrow{DB}$, c) $\overrightarrow{AB} + \overrightarrow{CD} + \overrightarrow{BC} + \overrightarrow{DA}$.

$$\begin{aligned}\vec{AB} + \vec{BC} + \vec{CD} &= \vec{AD} \\ \vec{AD} + \vec{BC} + \vec{DB} &= \vec{AB} + \vec{DC} = \vec{AB} \\ \vec{AB} + \vec{CD} + \vec{BC} + \vec{DA} &= \vec{AC} + \vec{CA} = \vec{0}\end{aligned}$$

13. Let $ABCD$ be a tetrahedron. Show that

$$\overrightarrow{AD} + \overrightarrow{BC} = \overrightarrow{BD} + \overrightarrow{AC}.$$

$$\begin{aligned}\vec{AD} + \vec{BC} &= \vec{BD} + \vec{AC} \Leftrightarrow \\ \vec{AD} + \vec{DB} + \vec{BC} + \vec{CA} &= \vec{0} \\ \vec{0} &= \vec{0}\end{aligned}$$

14. Let $SABCD$ be a pyramid with apex S and base the parallelogram $ABCD$. Show that

$$\overrightarrow{SA} + \overrightarrow{SB} + \overrightarrow{SC} + \overrightarrow{SD} = 4\overrightarrow{SO}$$

where O is the center of the parallelogram.

$$\begin{aligned}\vec{SO} &= \vec{SA} + \vec{AO} \\ \vec{SO} &= \vec{SB} + \vec{BO} \\ \vec{SO} &= \vec{SC} + \vec{CO} \\ \vec{SO} &= \vec{SD} + \vec{DO} \\ 4\vec{SO} &= \vec{SA} + \vec{SB} + \vec{SC} + \vec{SD} + \underbrace{\vec{AO} + \vec{BO} + \vec{CO} + \vec{DO}}_{\vec{0}} = \vec{0}\end{aligned}$$

15. Give the coordinates of the vertices of the parallelepiped whose faces lie in the coordinate planes and in the planes $x = 1, y = 3$ and $z = -2$.

$$\begin{aligned}O(0, 0, 0) & \quad H(0, 0, -2) \\ A(1, 0, 0) & \quad E(1, 0, -2) \\ B(1, 3, 0) & \quad F(1, 3, -2) \\ C(0, 3, 0) & \quad G(0, 3, -2)\end{aligned}$$

16. In E^3 consider the parallelograms $A_1A_2A_3A_4$ and $B_1B_2B_3B_4$. Show that the midpoints of the segments $[A_1B_1], [A_2B_2], [A_3B_3]$ and $[A_4B_4]$ are the vertices of a parallelogram.

$$\begin{aligned}\vec{C_1C_2} &= \vec{C_1B_1} + \vec{B_1B_2} + \vec{B_2C_2} = \\ &= \frac{1}{2}(\vec{A_1B_1} + \vec{A_2B_2}) + \vec{B_1B_2} \\ \vec{C_3C_4} &= \frac{1}{2}(\vec{A_3B_3} + \vec{A_4B_4}) + \vec{B_3B_4}\end{aligned}$$

$$\begin{aligned}\vec{C_1C_2} &= \vec{C_3C_4} \Leftrightarrow \\ (\vec{A_1B_1} + \vec{B_1B_2}) &= \vec{A_3B_3} + \vec{B_3B_4} \Leftrightarrow \\ \vec{A_1B_1} + \vec{B_1B_2} + \vec{B_2B_3} + \vec{B_3B_4} &= \vec{A_3B_3} + \vec{B_3B_4} + \vec{B_4B_1} + \vec{B_1B_2} \Leftrightarrow \\ \vec{A_1B_1} + \vec{B_2B_3} &= \vec{A_3B_3} + \vec{B_4B_1}\end{aligned}$$